

Blueprint: Dual-Scale String Theory and Double Field Theory in Lean 4

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Revision note

Revision 1 of this blueprint (“v1.0.0 Architecture Blueprint”) cited ten Lean declarations and marked all of them as formalized. Checking them against the compiled environment on 2026-09-17 showed that **eight of the ten names did not exist** (they had been renamed or never written), and several statements were described far more strongly than the Lean theorems prove — in particular the claims that “curvature singularities are mathematically forbidden by the Lean 4 kernel”, that the trans-Planckian censorship conjecture is satisfied “unconditionally”, and that the framework “mechanically predicts” cosmological and neutrino observables. The discrepancy had been recorded in `docs/RIGOR_ROADMAP.md` but not corrected.

This revision (i) replaces every reference by an existing declaration, (ii) states each result as the Lean statement reads, taken from `#check`, (iii) labels each claim by tier — [**Tier A**] kernel-checked, [**Tier L**] established in the literature, [**Tier C**] conjecture or proposal of this programme — and (iv) adds the `DualScaleStream2` library, whose results are genuine linear algebra over $\mathbb{Z}$ and $\mathbb{R}$. The older libraries cited in Chapters 1–5 model physical quantities by integers or small integer matrices: their theorems are kernel-checked, but they are *arithmetic models* of the physics, not formalizations of it. All theorems cited depend only on Lean’s standard axioms `propext`, `Classical.choice`, `Quot.sound`.

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Chapter 1

Double Field Theory and Generalized Geometry (integer models)

Definition 1.1 ($O(D, D)$ pairing). *Lean: DoubleFieldTheory.GeneralizedGeometry.ODD_Eta*

The pairing on $E = TM \oplus T^*M$ is $\eta = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. In this library it is a 2×2 integer matrix

(one generalized direction). The theorem

Lean: DoubleFieldTheory.GeneralizedGeometry.odd_metric_eval

(kernel-checked) states that evaluating the bilinear form of *ODD_Eta* on generalized vectors X, Y gives the pairing $\langle X, Y \rangle_\eta$. [**Tier A**] for this model.

Theorem 1.2 (Generalized metric, diagonal integer model). *Lean: DoubleFieldTheory.GeneralizedGeometry.gen_metric_duality*

(kernel-checked) For the diagonal integer matrix $\mathcal{H} = \text{diag}(g, g)$: if $g = 1$ then $\mathcal{H} \eta \mathcal{H} = \eta$.

Also

Lean: DoubleFieldTheory.GeneralizedGeometry.gen_metric_symmetric

(kernel-checked) (the two off-diagonal entries of $\text{diag}(g, g^{-1})$ agree) and

Lean: DoubleFieldTheory.GeneralizedGeometry.btwist_preserves_eta

(kernel-checked) (for the integer B -twist, the off-diagonal entries of $B^\top \eta B$ equal those of η).

[**Tier A**].

Scope. The general statement — $(\eta \mathcal{H}(G, B))^2 = 1$ for every invertible G and every B — is not proved in this library; it is proved in *DualScaleStream2*, Theorem 6.1.

Theorem 1.3 (C-bracket antisymmetry, component model). *Lean: DoubleFieldTheory.CourantAlgebroid.cbracket_antisymm*

(kernel-checked) For sections X, Y of the integer-component model *CourantSection*, the α -component of $[X, Y]_C$ is minus that of $[Y, X]_C$. [**Tier A**] for the model; antisymmetry of the C-bracket on doubled fields is [**Tier L**] (Hull–Zwiebach, arXiv:0904.4664).

Theorem 1.4 (Section condition, component model). *Lean: DoubleFieldTheory.CourantAlgebroid.strong_section_condition*

(kernel-checked) If the dual derivatives $\tilde{\partial}\Phi$ and $\tilde{\partial}\Psi$ vanish, then $\partial_M \Phi \partial^M \Psi = 0$ in the model. [**Tier A**] for the model.

Scope. That the strong constraint guarantees closure of generalized diffeomorphisms and reduces the DFT action to the NS–NS action is [**Tier L**] (Hohm–Hull–Zwiebach, arXiv:1006.4823). The Lean theorem *DoubleFieldTheory.ActionCurvature.dft_action_nsns_equivalence* states only the symmetry *ActionLagrangian a b = ActionLagrangian b a* of an integer model and does not formalize that reduction. For a genuine $O(d, d; \mathbb{Z})$ statement see Theorem 6.4.

Chapter 2

The Dual-Scale Bound

Definition 2.1 (Effective dual scale, integer model). *Lean: DualScaleValidation.UseCase1.effective_dual_scale_numerator*

For $R \in \mathbb{N}$ and $\alpha' = 1$, *effective_dual_scale_numerator* $R = R^2 + \alpha'$, the numerator of $R_{\text{eff}} = R + \alpha'/R$.

Theorem 2.2 (Lower bound, integer model). *Lean: DualScaleValidation.UseCase1.self_dual_is_global_minimum*

(kernel-checked) For every natural number $R \geq 1$: $R^2 + \alpha' \geq 2$. [**Tier A**].

Theorem 2.3 (Lower bound over the reals and for matrices). *Lean: DualScaleStream2.DualScale.circle_effective_scale_ge_two*

(kernel-checked) For every real $R > 0$: $2 \leq R + R^{-1}$. And

Lean: DualScaleStream2.DualScale.dualScale_ge

(kernel-checked) for every positive definite real $d \times d$ matrix G : $2d \leq \text{tr } G + \text{tr } G^{-1}$. [**Tier A**].

Scope. These are inequalities. In string units, $R + \alpha'/R \geq 2\sqrt{\alpha'}$ says that the T-duality-invariant effective scale of a circle is never below $2\sqrt{\alpha'}$ (the string scale, not the Planck length, as Revision 1 wrote). That this reflects a minimal physical length in string theory is [**Tier L**] (Giveon–Porrati–Rabinovici, arXiv:hep-th/9401139, around l. 792 of the text extraction); that it resolves cosmological curvature singularities is a proposal of this programme, [**Tier C**]. Revision 1’s sentence “singularities are mathematically forbidden by the Lean 4 kernel” is withdrawn: no Lean statement here concerns curvature or singularities.

Theorem 2.4 (Wavelength bound, integer model). *Lean: Lean5Corpus.Problems.DualScaleTCC.tcc_cosmic_protection_contract*

(kernel-checked) For natural numbers $R \geq 1$, $\lambda_0 \geq 1$ and any record p of inflation parameters: $(R^2 + 1)\lambda_0 \geq 2$; $(R^2 + 1)\lambda_0 \not\geq 1$ (the model’s Planck length); and $\text{planck_mass}/\text{hubble_scale} \geq 1$ for p — the last conjunct follows from the field $\text{hubble_scale} < \text{planck_mass}$ that every *InflationParameters* record must carry, i.e. it is assumed, not derived. [**Tier A**] for the model.

Scope. The Trans-Planckian Censorship Conjecture is a conjecture, [**Tier C**]. Nothing here shows that a cosmological model satisfies it; Revision 1’s “satisfied unconditionally” is withdrawn.

Chapter 3

Mathieu Moonshine Arithmetic

Theorem 3.1 (BPS ratio arithmetic). *Lean: DualScaleValidation.UseCase2.bps_cross_multiplication_lock*

(kernel-checked) With $\mathcal{A}_1 = 90$ and $\mathcal{A}_2 = 462$ (twice the Eguchi–Ooguri–Tachikawa coefficients $A_1 = 45$, $A_2 = 231$ of *arXiv:1004.0956*, eq. (1.12)): $462 \cdot 60 = 27720$ and $4 \cdot 90 \cdot 77 = 27720$.

Further

Lean: DualScaleValidation.UseCase2.bps_ratio_coprime

(kernel-checked) $\gcd(77, 60) = 1$, and

Lean: DualScaleValidation.UseCase2.m24_order_divisible_by_bps_lock

(kernel-checked) $|M_{24}| = 244\,823\,040 = 27720 \cdot 8832$. [**Tier A**].

Scope. These are arithmetic identities. That $A_1, \dots, A_5$ occur among the dimensions of irreducible representations of M_{24} is [**Tier A**] as a statement about the two tables (DualScaleStream2.Moonshine. that the ratio $462/(4 \cdot 90)$ is a physical “rigidity lock” is [**Tier C**].

Chapter 4

Tadpole Arithmetic on $K3 \times T^2$

Theorem 4.1 (D7/O7 charge bookkeeping, integer model). *Lean: `DualScaleValidation.UseCase3.rr_tadpole_cancellation`*

(kernel-checked) $total_d7_charge + total_o7_charge = 0$, where each total is the product of the brane (plane) count and the charge per brane (plane) fixed in the module. [Tier A] for these numbers.

Scope. That these numbers describe an orientifold of $K3 \times T^2$ as in Dasgupta–Rajesh–Sethi (arXiv:hep-th/9908088) is [Tier L]. The statement $\chi(K3 \times K3)/24 = 24$ computed from Hodge numbers is `DualScaleStream2.Flux.k3k3_anomaly`, Chapter 6.

Chapter 5

Observables (illustrative)

Theorem 5.1 (Window checks for chosen numbers). *Lean: DualScaleValidation. Observables. observables_windows_master_contract*

*(kernel-checked) For the numbers defined in the module — $r = 1/252$, $10^3 n_s = 965$, $\delta_{CP} = 283^\circ$, $w_0 = -1$, $w_a = 0$ — the following hold: $3 \cdot 252 \leq 1000 \cdot 1 \leq 5 \cdot 252$ (i.e. $0.003 \leq r \leq 0.005$), $950 \leq 10^3 n_s \leq 980$, $270 \leq \delta_{CP} \leq 295$, $w_0 = -1$, $w_a = 0$. [**Tier A**] as arithmetic.*

Scope. The Lean module itself calls the r and δ_{CP} windows “illustrative” and the value $r = 1/252$ “speculative”. No Lean statement derives these numbers from a compactification; they are [**Tier C**] proposals. Revision 1’s “the zero-parameter framework mechanically predicts” is withdrawn, and its value $\delta_{CP} = 282.4^\circ$ did not match the module (283°).

Chapter 6

DualScaleStream2: Lattices, $O(d, d; \mathbb{Z})$ and the Generalized Metric

This library (19 modules, 99 theorems, built on Mathlib) is the part of the formalization in which the objects are the actual mathematical ones: integer and real matrices of arbitrary size d .

Theorem 6.1 (The generalized metric squares to one against η). *Lean: DualScaleStream2. DFT. etaR_genMetric_sq*

(kernel-checked) For real $d \times d$ matrices G, B with $\det G$ a unit, $(\eta \mathcal{H}(G, B))^2 = 1$, where

$$\mathcal{H}(G, B) = \begin{pmatrix} G - BG^{-1}B & BG^{-1} \\ -G^{-1}B & G^{-1} \end{pmatrix}. \text{ [Tier A].}$$

Theorem 6.2 (B-shifts). *Lean: DualScaleStream2. DFT. genMetric_bshift*

(kernel-checked) If G is symmetric with $\det G$ a unit and B, Θ are antisymmetric, then

$$\Theta_R \mathcal{H}(G, B) \Theta_R^\top = \mathcal{H}(G, B + \Theta) \text{ for } \Theta_R = \begin{pmatrix} 1 & \Theta \\ 0 & 1 \end{pmatrix}. \text{ [Tier A].}$$

Theorem 6.3 (Duality group and spectrum). *Lean: DualScaleStream2. TDuality. isODD_mul*

(kernel-checked) $O(d, d; \mathbb{Z})$ is closed under products;
Lean: DualScaleStream2. TDuality. thetaShift_isODD
(kernel-checked) an integer Θ -shift with $\Theta^\top = -\Theta$ lies in $O(d, d; \mathbb{Z})$ (this direction only);
Lean: DualScaleStream2. TDuality. spectrum_equivalence
(kernel-checked) for $g \in O(d, d; \mathbb{Z})$ and any real $2d \times 2d$ matrix H there is a map e on integer charge vectors with $Z^\top (g^\top H g) Z = e(Z)^\top H e(Z)$ and e preserving the charge norm: dual backgrounds have the same (mass, level) spectrum. [Tier A].

Scope. That the exhibited elements generate all of $O(d, d; \mathbb{Z})$ is [Tier L] (GPR, arXiv:hep-th/9401139) and is not formalized.

Theorem 6.4 (The section condition is $O(d, d; \mathbb{Z})$ -covariant). *Lean: DualScaleStream2. DFT. isSection_image*

(kernel-checked) If $g \in O(d, d; \mathbb{Z})$ and S is a set of integer charge vectors that is a section (the η -pairing vanishes on all pairs from S), then the image $g(S)$ is a section. [Tier A].

Theorem 6.5 (Lattices). *Lean: DualScaleStream2. Lattice. cartanE8_posDef*

(kernel-checked) The E_8 Cartan matrix is positive definite over $\mathbb{R}$;
Lean: DualScaleStream2. Lattice. cartanE8_unimodular
(kernel-checked) it is unimodular (determinant ± 1);
Lean: DualScaleStream2. Lattice. sigK3_eq
(kernel-checked) adding the signature records $2 \cdot (0, 8) + 3 \cdot (1, 1)$ gives $(3, 19)$;

Lean: DualScaleStream2.Lattice.sigMukai_eq
(kernel-checked)adding one more $(1,1)$ gives $(4,20)$. [**Tier A**].

Scope. The two signature statements are additive bookkeeping: the inputs “ U has signature $(1,1)$ ” and “ $E_8(-1)$ has signature $(0,8)$ ” are definitions labelled Tier L in the source, not computed from the Gram matrices (positive definiteness of E_8 is proved separately by `cartanE8_posDef`). That $H^2(K3, \mathbb{Z})$ with its intersection form is isomorphic to $3U \oplus 2E_8(-1)$ is [**Tier L**] (Huybrechts, *Lectures on K3 Surfaces*).

Theorem 6.6 (Flux and moonshine). *Lean: DualScaleStream2.Flux.k3k3_anomaly*
(kernel-checked)for $\chi(K3 \times K3)$ computed from the Hodge-number table, $24 \mid \chi$ and $\chi/24 = 24$;
Lean: DualScaleStream2.Flux.flux_half_selfIntersection_integral
(kernel-checked)for symmetric integer Gram matrices L_1, L_2 with L_1 even, every integer vector x has $x^T(L_1 \otimes L_2)x$ even, so half the self-intersection of a flux in $L_1 \otimes L_2$ is an integer;
Lean: DualScaleStream2.Moonshine.first_five_are_irreps
(kernel-checked) $A_1, \dots, A_5$ occur in the M_{24} dimension table. [**Tier A**].

Reproduce. lake build DualScaleStream2 DualScaleValidation DoubleFieldTheory Lean5Corpus,
then python3 tools/axiom_audit.py <library> for each. The certificate with verbatim
statements is docs/VERIFIED_FOUNDATION.md.